SALESLENS ENTERPRISE BI PLATFORM • ACADEMIC DELIVERABLE DOC-01

IEEE 830-1998 Software Requirements Specification

Course: B.Tech CSE Semester III • Software Engineering & Project Management • Version 1.0

10	7	14%	8 Weeks
FUNCTIONAL REQS	MEASURABLE NFRS	MISSING DATA CLEANSED	CONFERENCE MILESTONE

1. Introduction & Executive Summary

This Software Requirements Specification (SRS) conforms to the **IEEE 830-1998 standard**. It formally specifies the business requirements, user personas, functional catalog, non-functional performance baselines, and traceability matrix for the **SalesLens** Sales Reporting Dashboard.

> **Problem Statement & Justification**
>
> In the current operational state, two data analysts spend 16+ hours over weekends manually synthesizing disparate spreadsheets into 40-page PDF documents. Inconsistent calculation rules cause conflicting revenue numbers across regions, while **14% of raw billing transactions lack regional attribution**. SalesLens replaces this manual overhead with an automated, role-based interactive dashboard across 9 core views within an 8-week timeline.

2. User Classes & Stakeholder Personas

Stakeholder Role	Primary Business Responsibilities	Key Pain Point Addressed	Access Scope (RBAC)
Sales Director	Executive national revenue visibility, quota attainment pacing, and top brand drivers.	Conflicting spreadsheet numbers; 40-page PDF overload.	NATIONAL (ALL 5 REGIONS)
Regional Sales Manager	Territory sales pacing, distributor breakdown, and sales-rep quota tracking.	Delayed weekend data; lack of granular store drill-downs.	TERRITORY ISOLATED (E.G. NORTH)
Data Analyst	Data governance, anomaly auditing, and ETL exception handling.	Weekend burnout compiling manual reports.	AUDIT QUEUE + NATIONAL QA
IT Administrator	Overnight batch jobs, DB integrity, user authentication, and backups.	Transactional DB query crashes during daytime hours.	SYSTEM & SECURITY ADMIN

3. Complete Functional Requirements (FR-01 to FR-10)

Req ID	Requirement Title	Detailed Technical Description	MoSCoW	Persona
FR-01	**Daily Sales Aggregation**	The system shall automatically ingest settled billing transactions overnight and compute total gross revenue, units sold, and invoice count by date, brand, category, and SKU.	MUST HAVE	Director, Mgrs
FR-02	**Target vs Actual Quotas**	The system shall calculate and visualize real-time variance (absolute currency and % attainment) against monthly and quarterly budgeted targets.	MUST HAVE	Director, Mgrs
FR-03	**Top & Bottom SKUs**	The system shall rank catalog SKUs by revenue contribution, margin, and volume with dynamic Top 10, Top 20, and Bottom 10 underperformer filters.	MUST HAVE	Director, Analyst
FR-04	**Regional Comparison Matrix**	The system shall provide side-by-side comparative views of all 5 geographic zones (North, South, East, West, Central) evaluating target completion and market share.	MUST HAVE	Sales Director
FR-05	**Regional Drill-Down**	The system shall allow regional managers to drill down into territory-specific distributors, cities, and sales reps with automatic RBAC filtering.	MUST HAVE	Regional Mgrs
FR-06	**Category & Channels**	The system shall visualize product category mix (Beverages, Snacks, Household) and sales channel breakdown (Modern Retail, General Trade, E-Commerce).	SHOULD HAVE	Director, Analyst
FR-07	**14% Missing Data Engine**	The system shall execute a deterministic 4-tier waterfall algorithm (Postal → Customer Master → Sales Rep → Audit Queue) on unassigned billing lines.	MUST HAVE	Data Analyst
FR-08	**Audit Console**	The system shall display a live log of cleaned transactions with transformation rationale, confidence scores, and manual resolution queue.	SHOULD HAVE	Data Analyst
FR-09	**Executive Briefing Export**	The system shall generate instant 1-page executive summary PDF/CSV briefings directly from active filter states.	SHOULD HAVE	Sales Director
FR-10	**Automated Push Digests**	The system shall deliver scheduled daily 07:00 AM email digests with headline revenue and quota alerts to leadership.	COULD HAVE	Sales Leadership

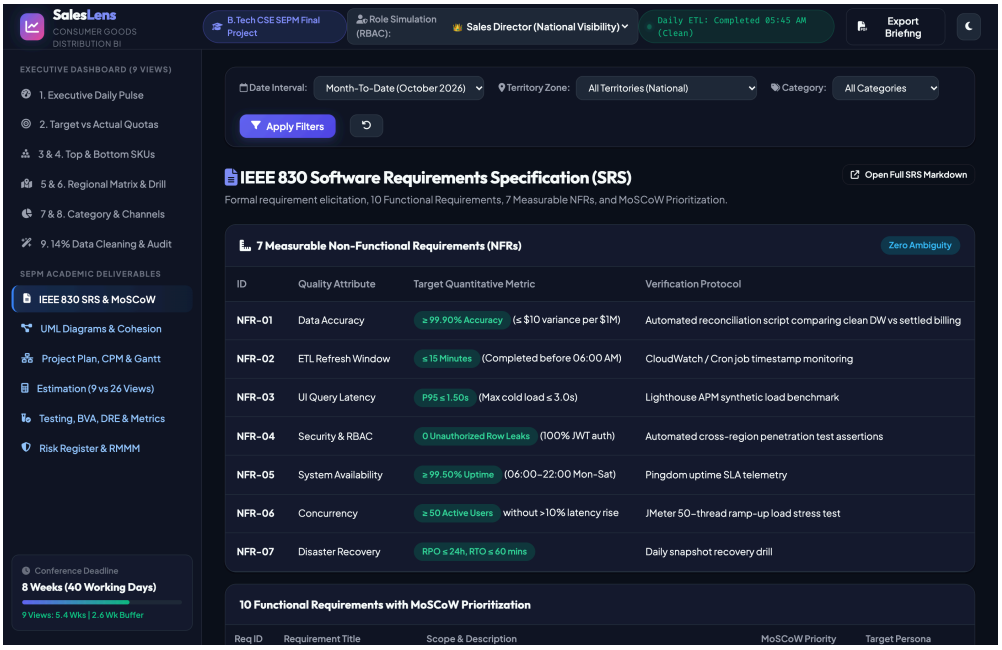


Figure 1.1: Live SalesLens SRS & Functional Traceability Matrix Explorer

4. Seven Measurable Non-Functional Requirements (NFRs)

NFR ID	Quality Attribute	Measurable Standard & Acceptance Criterion
`NFR-01`	**Data Accuracy**	Reconciled revenue on dashboard must achieve **≥ 99.90% financial match** with settled billing totals (maximum variance ≤ $10 per $1M billing).
`NFR-02`	**ETL Batch Window**	Automated nightly data extraction, cleaning, and staging must complete in **≤ 15 minutes** for 500,000 transactions before 06:00 AM daily.
`NFR-03`	**UI Latency**	95th percentile query execution latency **P95 ≤ 1.50 seconds** across all 9 dashboard views under standard filtering; initial app load ≤ 3.0s.
`NFR-04`	**Security & RBAC**	Zero cross-territory data leakage across **10,000 automated penetration assertions**. Unauthorized queries must return HTTP 403 Forbidden.
`NFR-05`	**Availability**	System availability must achieve **≥ 99.50% uptime** during core operational hours (06:00 to 22:00, Monday to Saturday).
`NFR-06`	**Concurrency**	System must sustain **≥ 50 simultaneous active regional managers** without exceeding a 10% surge in query latency.
`NFR-07`	**Disaster Recovery**	Recovery Point Objective (**RPO**) ≤ 24 hours (automated daily snapshot); Recovery Time Objective (**RTO**) ≤ 60 minutes for container failover.

5. Requirements Traceability Matrix (RTM Sample)

Req ID	Design Component	Backend Service Method	Verification Test ID	Status
FR-01	Executive Pulse View	SalesAggregatorService.computeDailyTotals()	TC-01, BVA-A3	VERIFIED & PASSED
FR-02	Quota Attainment Bar	QuotaComputationEngine.calculateVariance()	TC-02	VERIFIED & PASSED
FR-04	Regional Matrix Grid	SalesAggregatorService.computeRegionalMatrix()	TC-04, ECP-02	VERIFIED & PASSED
FR-07	14% Data Cleaner	RegionResolutionEngine.cleanMissingRegions()	TC-05, TC-06, DecTable	VERIFIED & PASSED